

precision agriculture are given below.

1. Location sensors use signals from GPS satellites to determine latitude, longitude and altitude within the required area. A minimum of three satellites are required to triangulate a position. NJRNJG1157PCDTEI GPS integrated circuit (IC) is a good example of location sensors.

2. Optical sensors use light to measure soil properties. These measure different frequencies of light reflectance in near-infrared (IR), mid-IR and polarised light spectra. Sensors can be placed on vehicles or aerial platforms, such as drones or satellites, to determine clay, organic matter and moisture content of the soil.

3. Electro-chemical sensors provide key information in the form of pH and soil nutrient levels. Sensor electrodes work by detecting specific ions in the soil. Currently, sensors mounted on specially-designed sleds are helping gather, process and map soil's chemical data.

4. Mechanical sensors measure soil compaction or mechanical resistance.

5. Dielectric soil moisture sensors assess moisture levels by measuring dielectric constant of the soil.

6. Air flow sensors measure soil air permeability. Measurements can be made at a single location or dynamically while in motion. Desired output is the pressure required to push a predetermined amount of air into ground at prescribed depth. Various types of soil properties, including compaction structure, soil type and moisture level, produce unique identifying signatures.

Agricultural weather stations. These are self-contained units placed at various locations throughout the agricultural field. These stations have several sensors that are appropriate for local crops and climate. Information such as air temperature, soil temperature at various depths, rainfall, leaf wetness, chlorophyll, wind speed, dew point temperature, wind direction, relative humidity, solar radiation and atmospheric pressure are measured and recorded at predetermined levels. This data is analysed and sent wirelessly to a central data server at programmed intervals. The stations

are portable and less expensive, and available at attractive prices for farms of all sizes.

Smartphone tools. A number of smartphone tools can be adapted to farming applications. For instance, crop and soil observations can be logged in the form of pictures, pinpointing locations, soil colours, water, plant leaves and light properties.

Tools such as cameras, GPSes, microphones, accelerometers, gyroscopes and smartphone applications can greatly help farmers. Using these, they can identify crop diseases and make diagnoses. They can even calculate the amount of fertiliser required, and study the soil and water.

In the developing world, roughly 500 million small farms produce more than 80 per cent of the total food consumed. Agricultural sensors are becoming more widely accessible around the globe. According to UN projection, worldwide demand for food will increase by 50 per cent by 2050, and as such, precision agricultural technologies for farmers will be in demand.

Crop breeders, researchers and representatives from universities and industries are discussing new and improved ways of breeding resistant and highly productive crops.

Harvests will have to expand to feed a growing world population over the next decades. To achieve the goal without expanding cultivated areas at the expense of the environment, agricultural economists agree that agricultural sensors can greatly assist in this direction.

Uses of agricultural sensors in India

Several agricultural sensor-based equipment have been developed by agro-based electronic industries. For example, Waspnote Agriculture Board developed in India allows monitoring of multiple environmental parameters involving a wide range of applications. For this, it has been provided with sensors for measuring air and soil temperature, humidity, solar radiation, wind speed and direction, rainfall, atmospheric pressure, leaf wetness, and fruit or trunk diameter.

Up to 14 sensors can be con-

nected at the same time to a dendrometer (device used to measure trees). To extend durability of the device after deployment, the equipment is endowed with a solid-state switch, which facilitates precise regulation of its power, prolonging the life of the battery.

Agricultural sensors are slowly taking foot in India. Vertical farming is becoming popular in smart cities. Drones are being used in large-scale farming for insecticide and pesticide spraying purposes. However, these sensor-based devices are yet to become popular in India. These need government assistance and large subsidies in prices.

Indian Space Research Organization (ISRO) has developed an Android-based application that collects real-time information to assess the damage caused to agricultural crops due to hail storms. This application allows the farmers to process faster insurance claims. It is currently used for rice and cotton crops in the states of Karnataka, Madhya Pradesh, Haryana and Maharashtra.

India faces various challenges in applying smart farming and the Internet of Things (IoT) in agriculture. Internet connectivity and availability is a major challenge. Then, there are aspects like presumption among vendors about Indian customers not being ready for advanced products. This results in drastically low awareness regarding IoT devices and systems among consumers. Also, the infrastructure required for the smooth functioning of IoT devices, such as smart grids and traffic systems, are far from ready. This further hinders the growth.

Among the technologies invented in the past few decades, smartphones have gained the largest market share in the Indian agricultural industry. This is due to their usefulness, ease of use and affordability. Rural farmers in India are constrained by a low level of literacy and lack of exposure to software interfaces. Therefore it is necessary for smartphone application developers to target farmers while developing agricultural applications.

Mobile-operated solar-based pumps reduce the cost of electricity